The Rodin coil in quadrature

A counter-wound Rodin pair driven by a sine and a cosine: its field, its resonance, the winding as a line on a knot, and output coils charging a capacitor bank — register entry H43

kronspice simulation report · 2026-11-08 · GA seeds 43 and 44, 64 × 50

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Summary

The Rodin coil winds its wire over a torus and through the hole, 150° round the axis per pass between 12 poles: 1→6→11→4→9→2→7→12→5→10→3→8→1. Seen from above it is a {12/5} star; in space one circuit goes five times round the axis and twelve times through the hole, a torus-knot path of the (5, 12) class. This example builds the coil characterized by Ramírez Galindo et al. (*Rev. Mex. Fís.* 70, 030901, 2024) as a counter-wound pair, drives the windings with a sine and a cosine, maps the field, builds it up by resonance, and evolves output coils, together with the drive frequency, that charge a capacitor bank. Every joule is booked against the two drive sources.

- **Validated against the paper:** one fitted turn count (12 circuits per winding of 0.6 mm wire) gives 3.948 and 3.084 mT at 1.60 and 1.25 A against 4.13 and 3.05 mT measured (−4.4 %, +1.1 %). At 0.60 A the model gives 1.480 mT against 1.16: the paper's own data are not linear in the current.
- **Sine and cosine turn the field vector but do not move the pattern:** ellipticity 0.09–0.11 in the planes, while the 12-fold pattern round the axis advances by at most 3×10^{-13}° per cycle.
- **The sources pass power through the coupling:** 0.4401 W from A to B (toroidal sense) and 0.4546 W from B to A (dipole sense), equal to $\omega\cdot M\cdot I_A\cdot I_B\cdot\sin\varphi$ to four figures.
- **Energy accumulates only by resonance:** a tuned winding rises to its Q-limited level, Q = 76.4 from the rise against $\omega_0 L/R$ = 77.3, paid by the source. With k = 0.39 the pair cannot be tuned as a pair; tuned to one mode its currents lock in phase (−0.2°) under a sine/cosine drive.
- **A line on a knot near 2 MHz:** first self-resonance 1.966 MHz; the series counter-wound loop closed on itself is photon_electron's crossed ring, with 720° modes for the (5, 12) and the (2, 1) path alike.
- **Output coils:** the free search builds a transformer (one coaxial coil, 4.1 kHz; bank 0.176 of the source energy); two coils under the aperture at 29 kHz charge the bank with 0.321. The books close to the transient's integration error. Status: **bounded**.

What is being tested

Hypothesis (H43). Hobbyist and promotional sources (M. Rodin; forum posts) say the Rodin winding makes a “vortex” field that accumulates energy, delivers more than its input, and follows a “vortex-based mathematics” (the doubling sequence 1-2-4-8-7-5). No independent measurement is given, and the mathematics gives no equation that a simulation could check.

Standard prediction. Two coupled inductors: a transformer, and near self-resonance a transmission line. The field is the Biot–Savart field of the currents; energy builds up only in a resonance, to the level where loss per cycle equals input; output is at most input, and the ledger closes.

The relation to the photon electron (H23)

examples/photon_electron tests Williamson and van der Mark's electron, one photon of the Compton wavelength on a (2, 1) torus-knot path whose field returns to itself only after 720°. It builds that property as a circuit — a two-wire ring line with its conductors crossed once, whose modes are half-integer and whose pulse returns inverted after one trip — and adds the circuit mirror of the fdamador neutrino: the neutrino's field (E round the ring, B through the hole) is a series RLC loop, and the electron's mirror is a parallel RLC tank.

The Rodin winding is the same family of paths with other winding numbers, (5, 12) instead of (2, 1). It carries a current in a wire, not light, so the wire does the confining. At the drive frequencies used here (1 kHz, 100 kHz) the

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19.2 m of wire per winding is far shorter than a wavelength, so the winding is a lumped inductor. Only near its self-resonance (1.966 MHz) does it become a transmission line laid on a knot. The counter-wound pair connected in series and closed on itself *is* the crossed ring: its lowest modes come back inverted after one trip through one winding. They do so for the (5, 12) Rodin winding and for a (2, 1) winding on the same former alike, so the 720° property belongs to the connection, not to the knot's winding numbers. In the mirror language, a winding tuned by a series capacitor is the series RLC loop.

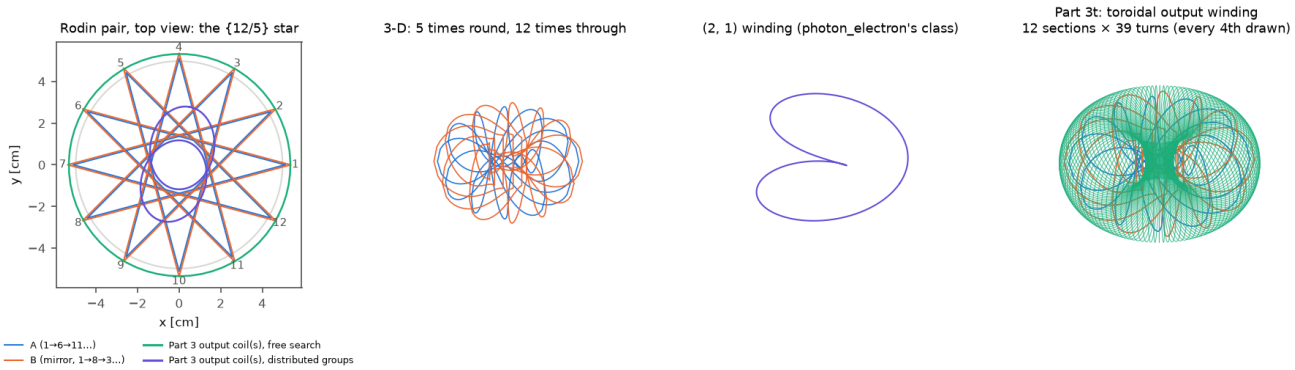


Figure 1. Left: the Rodin pair from above — winding A (1→6→11…) and its mirror B, the 12 poles, and the Part 3 output coils beside the toroid (green: free search; violet: distributed groups). Middle: one circuit of each winding in 3-D. Third: a (2, 1) winding, the class of photon_electron's path. Right: the Part 3t toroidal output winding, 12 sections of 39 turns, every turn round the tube through the hole.

Part 1. Geometry and validation

Taut-wire path: from above, each pass is the chord from pole P_k to P_{k+5}, the first half on the top surface of the torus and the second underneath, changing side through the hole. Former R_0 = 3.2125 cm, a = 1.7625 cm. The paper gives no turn count or gauge: 0.6 mm copper is declared, each winding lays its path Nc times in a round bundle, and Nc is fitted to the measured centre field (A and B in series, the paper's connection): Nc = 12, 2.4672 mT/A.

I [A]	paper theory	paper FEM	measured	kronspice	vs measured	vs theory
1.60	3.90	4.08	4.13	3.948	−4.4 %	+1.2 %
1.25	3.01	3.00	3.05	3.084	+1.1 %	+2.5 %
0.60	1.30	1.10	1.16	1.480	+27.6 %	+13.9 %

check	result
5 cm ring vs Kirchhoff $\mu_0 R(\ln(8R/r) - 7/4)$	3.42106 vs 3.42147 $\times 10^{-7}$ H (-1.2×10^{-4})
L_A, L_B, M, k	263.44 µH, 310.53 µH, +111.97 µH, 0.3915
series inductance, toroidal / dipole sense	797.91 / 350.03 µH
reciprocity M_{AB} vs M_{BA}	111.9694754 µH both (2.4×10^{-16})
flux of A's field through a 1 cm circle vs Neumann	516.443 vs 516.388 nH (1.1×10^{-4})
the same, circle round the tube (3 mm from the bundles)	−1123.68 vs −1090.02 nH (3.1 %)
field energy outside a 2.5 mm tube vs $\frac{1}{2}L_{ext}I^2$, (2, 1) circuit	125.02 vs 125.7 nJ (−0.54 %)
the same, one Rodin circuit (tubes overlap at crossings)	695.14 vs 743.79 nJ (−6.54 %)
wire resistance A / B	1.1414 Ω (19.209 m) / 1.3212 Ω (22.235 m)

Part 2. Sine on A, cosine on B

Lumped two-winding circuit in kronspice, sources 3.383 V peak at 1 kHz through 1 Ω (A alone carries 1.25 A). **Toroidal sense:** the through-the-hole parts of A and B add (field in the tube). **Dipole sense:** the round-the-axis parts add (field through the hole, the paper's connection).

drive	$\lvert I_A\rvert$ A	$\lvert I_B\rvert$ A	P_A W	P_B W	A→B via M, W	$\omega M I_A I_B \sin\varphi$	W_m µJ
A only	1.2499	0	1.6728	0	0	0	0.022–205.8
B only	0	1.1156	0	1.4446	0	0	0.026–193.2
in phase, toroidal	1.0706	0.9521	1.2544	1.0253	−0.0270	−0.0268	0.26–405.3

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drive	$\lvert I_A \rvert$ A	$\lvert I_B \rvert$ A	P_A W	P_B W	A→B via M, W	$\omega M I_A I_B \sin\varphi$	W_m µJ
in phase, dipole	1.4327	1.2944	2.1860	1.9569	+0.0126	+0.0122	0.07–322.9
sin/cos, toroidal	1.4624	0.8908	1.8500	1.3613	+0.4401	+0.4401	104.3–260
sin/cos, dipole	1.0297	1.3373	1.5904	1.6209	−0.4546	−0.4548	117.1–247.2

Every case closes its books over the last cycle (sources = copper + source resistors to ≤ 1.6×10^{-5} W) and over the run (ledger mismatch ≤ 9×10^{-7} of the energy in).

drive	centre peak mT	tube, pole: peak mT (ellipticity)	largest peak z=0 / y=0 mT	mean $B^2/2\mu_0$ z=0 / y=0 J/m³	ellipticity z=0 / y=0
A only	1.9339	1.1292 (0.000)	2.2098 / 2.1852	0.1137 / 0.06669	0 / 0
in phase, toroidal	0.7860	1.8229 (0.002)	3.5089 / 3.2838	0.2606 / 0.08905	0.003 / 0.007
in phase, dipole	3.4075	0.1892 (0.090)	3.4925 / 3.6579	0.05043 / 0.1572	0.008 / 0.002
sin/cos, toroidal	2.6128	1.3403 (0.065)	2.8371 / 2.8258	0.1676 / 0.1249	0.090 / 0.094
sin/cos, dipole	2.3243	1.2433 (0.078)	2.4479 / 2.7285	0.1434 / 0.1214	0.108 / 0.098

The quadrature pair: the toroidal and dipole parts of the field peak a quarter cycle apart, so the field at a point traces an ellipse; the pattern itself keeps the poles' 12-fold symmetry at every instant and does not travel round the axis. A linear coil's field is the superposition of the drives at each instant; nothing accumulates from cycle to cycle without a resonance.

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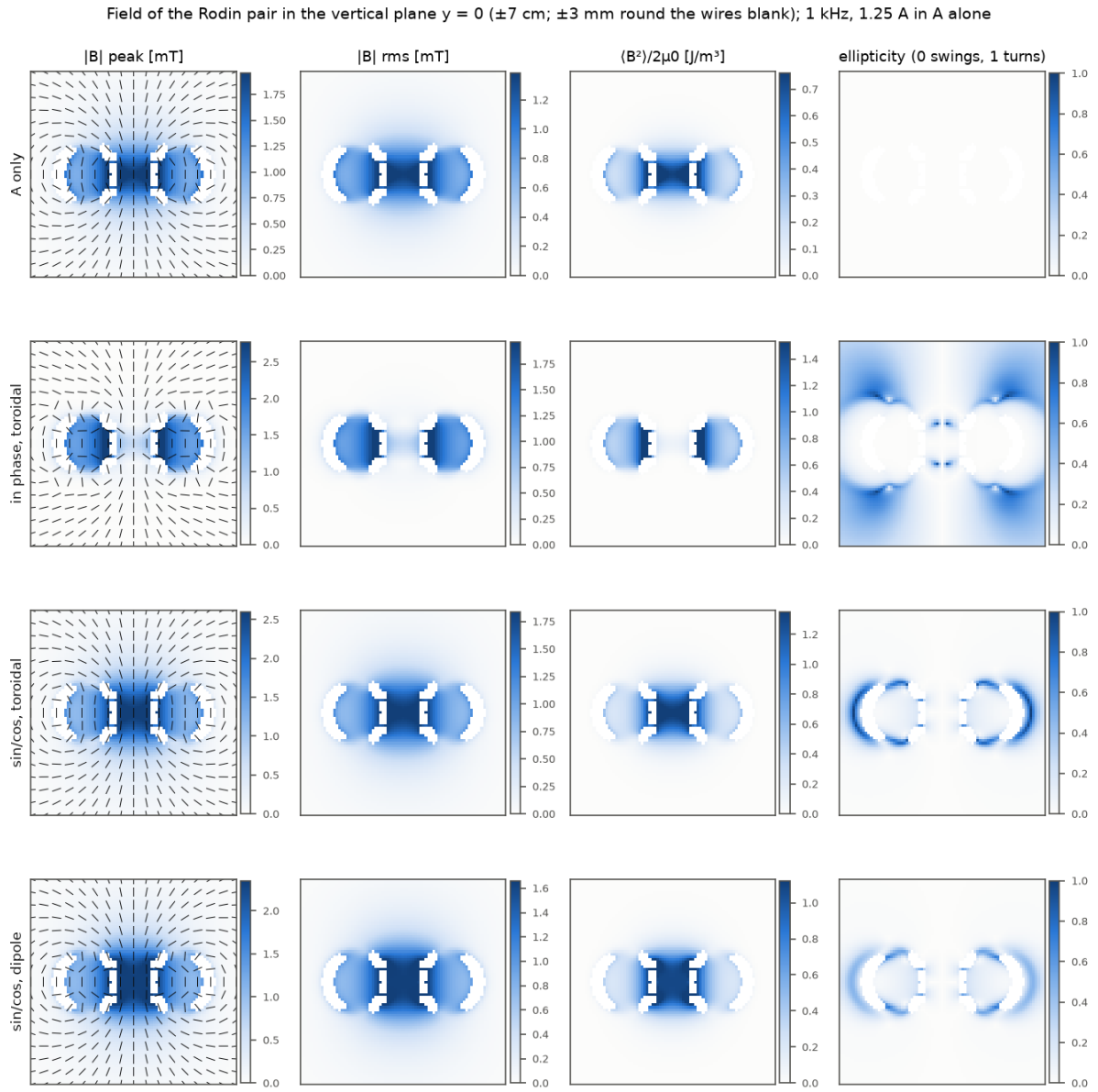


Figure 2. The field in the vertical plane y = 0 (±7 cm) for four drives: |B| peak over a cycle with the direction of the ellipse's major axis, rms, time-averaged energy density, and ellipticity (0: the field only swings; 1: it turns at constant size). viz/fields_xy.png shows the mid-plane; viz/cycle.gif animates one cycle of the sine/cosine drive.

case at 100 kHz	$\|I_A\|$ A	$\|I_B\|$ A	$\varphi_B - \varphi_A$	mean W_m μJ	P_{in} W	$Q = \pi f_0 \tau$	$\omega_0 L/R$
A only, tuned to L_A	1.2480	0	—	102.6	1.6686	76.4	77.3
A only, not tuned	0.0162	0	—	0.0172	0.0003	—	—
sin/cos, toroidal, each tuned to own L	0.0392	0.0368	−90.2°	0.2065	0.0032	—	—
sin/cos, toroidal, not tuned	0.0204	0.0175	132.8°	0.03753	0.0008	—	—
in phase, toroidal, tuned to L + M	1.1957	1.1958	−0.1°	285.2	3.1930	111	—
sin/cos, toroidal, tuned to L + M	0.8594	0.8327	−0.2°	142.5	1.5969	112	—

One tuned winding builds up over τ = 2L/R to the level where loss per cycle equals input: 5,965 times the untuned winding's stored energy at the same voltage, Q = 76.4 against 77.3 predicted. With k = 0.39 the pair has two modes, L ± M; tuning each winding to its own L leaves the drive between them, and tuned to the L + M mode the currents lock in phase under the sine/cosine drive: a resonant pair carries its mode, and the turning vector is gone.

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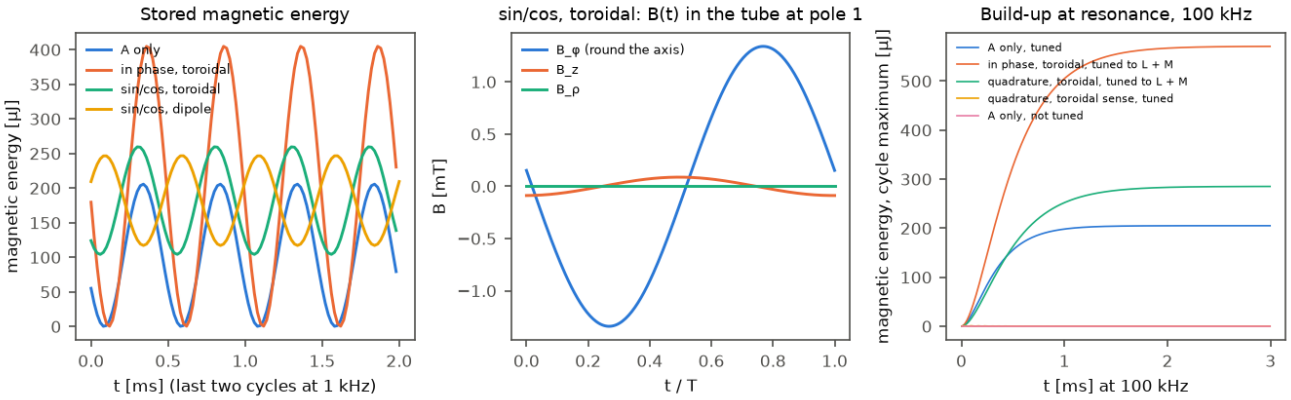


Figure 3. Left: stored magnetic energy over two cycles at 1 kHz. Middle: B(t) in the tube at pole 1 under the sine/cosine drive (toroidal sense). Right: the build-ups at 100 kHz (cycle maxima).

Part 2b. The winding as a line on a knot

Sectional model: every pass of every circuit is a section (144 per winding), the inductance matrix from the geometry, declared capacitances (188.1 pF/m between neighbouring wires of a bundle, A–B contact ×0.1, 5.54 pF to the surroundings). Lossless eigenmodes; in brackets the trip parity (−1: inverted after one trip through one winding, the 720° modes).

winding	open (far end open)	series, toroidal	series, dipole	plain rings	crossed ring
Rodin (5, 12), 19.209 m	1.966, 13.66 MHz	0.5314, 1.266	0.7052, 1.578	9.534 (−1.00), 9.814 (−1.00)	1.27 (−1.00), 6.258 (−1.00)
(2, 1) knot, 5.091 m	8.039, 50.51 MHz	3.578, 4.091	2.279, 10.34	38.85 (−1.00), 40 (−1.00)	4.144 (−1.00), 25.2 (−1.00)

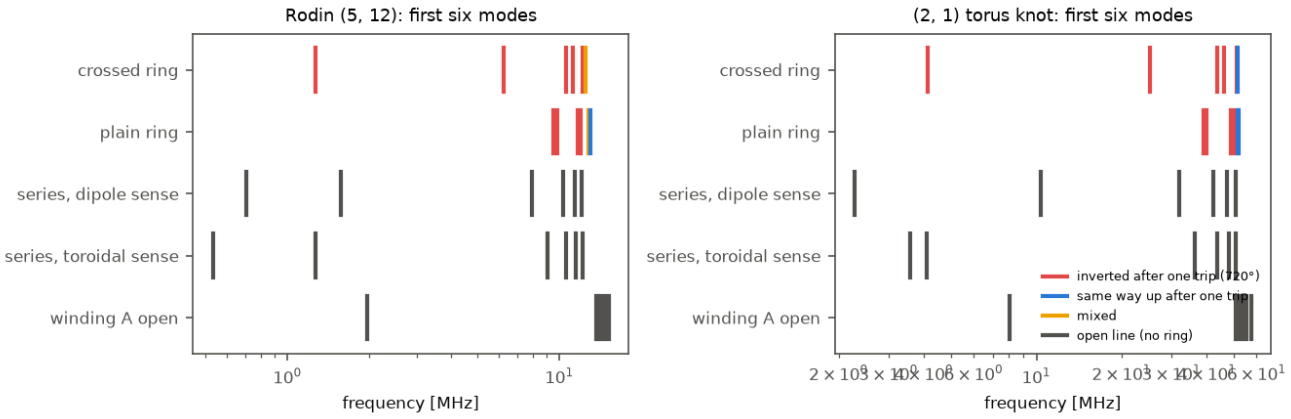


Figure 4. The first six modes of each network, coloured by trip parity.

Part 3. Output coils round the toroid, the frequency evolved with them

Sine on A, cosine on B, 3.383 V peak each. Output coils of 0.5 mm copper, each tuned to the drive frequency by a series capacitor, coupled to every section of the pair through mutuals computed from the geometry; they load the drive in every solution. The sectional networks are solved by a complex modified-nodal solver written for the example (mutuals of either sign), twinned against a kronspice transient of the same network to 3×10^{-3}.

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Where round the toroid, at which frequency

coil, sense	EMF max/min over angle: 1 kHz	100 kHz	1.86 MHz	5.28 MHz	18.9 MHz	largest load power
tube, toroidal	1.04 (m 12)	1.05 (m 12)	1.09 (m 12)	4.13 (m 1)	26.4 (m 5)	0.03586 W, 5.68 kHz
tube, dipole	1.05 (m 12)	1.05 (m 12)	1.43 (m 2)	112 (m 5)	5.01 (m 1)	0.03851 W, 4.51 kHz
top, toroidal	1.02 (m 12)	1.05 (m 12)	2.51 (m 5)	106 (m 2)	64.9 (m 7)	0.03318 W, 18.9 MHz
top, dipole	1.07 (m 12)	1.04 (m 12)	1.24 (m 3)	12.5 (m 7)	57 (m 7)	0.0331 W, 18.9 MHz

Below the first self-resonance every pass carries the same current, so the field keeps the 30° symmetry of the poles and a coil sees the same EMF within 2–7 % at every angle (harmonic m = 12): where a group sits round the toroid does not matter, and the sine/cosine pair gives no angle-dependent phase, since the pattern does not turn. Above it the current varies along the wire as a standing wave; the wire goes round the axis 5 times per circuit, 150° per pass, so the field round the toroid takes the low harmonics of that wave (m = 1, 2, 5, 7) and the EMF changes by up to 112× with angle, the best angles moving with frequency. These modes depend on the declared capacitances.

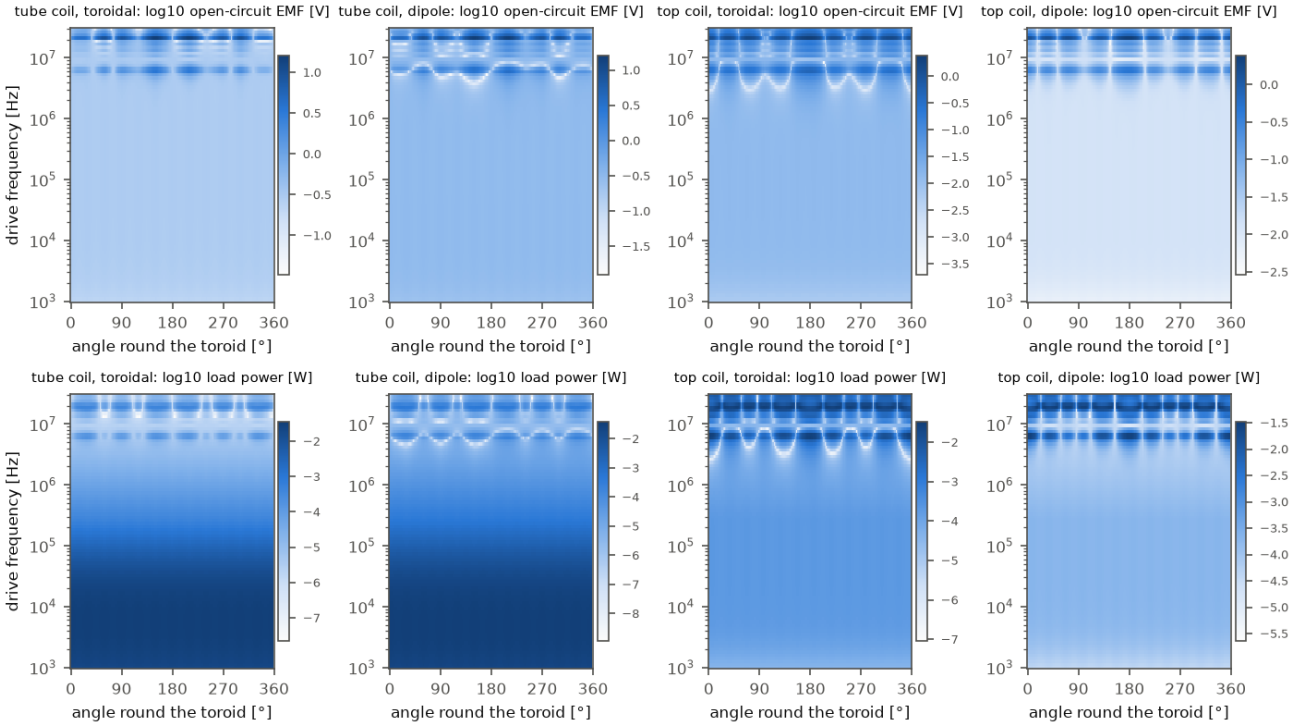


Figure 5. $\log_{10}$ of the open-circuit EMF (top, full sectional net) and of the load power (bottom, coil in the network, reduced net) of a reference coil against the drive frequency and its angle round the toroid; tube coil and top coil, both connection senses.

The genetic searches

Genes: drive frequency (1 kHz–30 MHz), number of groups (1–8), first angle and spacing, coil tilt, position, radius (3–60 mm), turns (1–200), combination (each into its own bridge, all in series into one, all in parallel into one), load, connection sense. Constraints: coils clear of the bundles by 2 mm and of each other, shorter than λ/20, tuning capacitors ≥ 30 pF. Fitness: load power in the AC solution times a smooth bridge factor $(V/(V + 0.6\ V))^2$. Searches run on 2 sections per circuit; the winners give the same power on 4, 6 and 12.

	free search	distributed groups (≥ 2, radius ≤ 20 mm)
fitness, generation 0 → evolved	1.664×10^{-7} → 0.181 W	1.265×10^{-7} → 0.1172 W
drive frequency, sense	4,134 Hz, dipole	29.38 kHz, dipole
groups	1 coaxial coil round the whole coil, r 53.3 mm, 200 turns, 14.4 mm above the mid-plane	2 at 79.7° and 251.1°, r 20 mm, 200 turns, tilted 50.1°, under the aperture (z −34.8 mm), in series
coil L, R, tuning C	7677 µH, 5.733 Ω, 193.1 nF	1894 µH, 2.15 Ω, 15.5 nF
AC: sources = load + copper + source R	0.4708 + 0.2793 = 0.2522 + 0.307 + 0.1909 W	0.2061 + 0.1462 = 0.2018 + 0.126 + 0.02447 W

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	free search	distributed groups (≥ 2, radius ≤ 20 mm)
reaction: sources without → with the coils	0.5877 + 0.03219 → 0.4708 + 0.2793 W	0.058 − 0.04325 → 0.2061 + 0.1462 W
separate / series / parallel bridges	one coil: 0.2522 W	0.1598 / 0.2018 / 0.1053 W

Left free, the search builds a transformer: one coil coaxial with the toroid, linking the dipole flux through the hole. Made to use several small coils, it puts two under the aperture where that flux leaves, in phase (138.3° both) and in series; in parallel they give half as much, on separate bridges 21 % less. Neither search goes above the self-resonance: the angle matters there, but the winding's input impedance leaves less power to take.

The winners charging the bank

kronspice transient	free search	distributed groups
run, bank	400 cycles (96.77 ms), 2200 µF	400 cycles (13.62 ms), 680 µF
bank voltage at the end	3.367 V	1.86 V
bank energy / source energy A + B	12.47 mJ / 47.04 + 23.65 mJ	1.176 mJ / 2.393 + 1.266 mJ
charging efficiency	0.1764	0.3214
copper / source R / diodes	45.08 / 28.96 / 6.47 %	34.76 / 8.10 / 21.65 %
books: sources − (bank + losses + stored)	35.6 µJ	4.95 µJ
ledger integration mismatch	35.8 µJ (4.0×10^{-4})	4.97 µJ (1.1×10^{-3})
radiation of the output coil	3.59×10^{-15} Ω vs 5.73 Ω	1.82×10^{-13} Ω vs 2.15 Ω

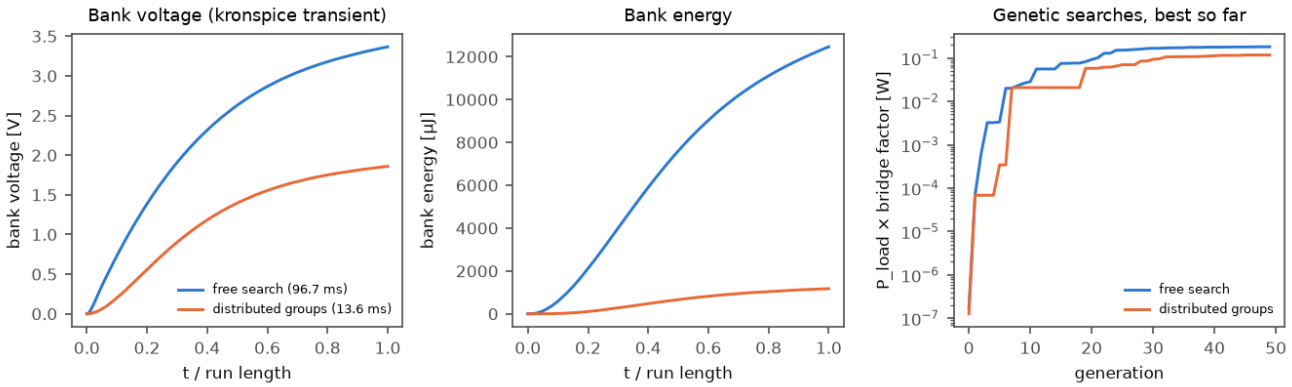


Figure 6. Bank voltage and energy of the two winners in the kronspice transient (Schottky bridges), and the best fitness of each genetic search by generation.

Part 3t — a toroidal output winding in 12 sections

The output winding wound round the torus as on a toroidal inductor core: every turn over the outside, through the hole and back under, laid on the two Rodin windings, in 12 sections 30° apart (arc ≤ 29°, 0.5 mm copper, up to 6 layers; a 29° section holds 39 turns). Each turn's mutual to every Rodin segment is tabulated over 30° and carried round by the windings' 30° symmetry (pass q of A → q + 5, of B → q + 7). Checks: interpolation 1.6×10^{-4}; a uniform 114-turn toroid 158.59 vs the current sheet 158.76 µH; rings vs an explicit helix with its return lead, mutuals 0.25 %, self 3 %.

A toroidal turn collects only the toroidal flux of the through-the-hole parts. Under sine/cosine the two windings' parts are a quarter cycle apart, so a turn sees $\sqrt{(A^2 + B^2)}$ in either sense (11.10 mV per turn at 1 kHz); driven in phase in the toroidal sense they add, A + B (15.65 mV). Round the torus the EMF varies by 3–4 % (m = 12) below the self-resonance, so where the sections sit hardly matters (0.16 % in load power).

	sine on A, cosine on B	drive phase evolved too
frequency, phase of B, sense	1,575 Hz, +90°, dipole	1,557 Hz, 356.4° (in phase), toroidal
winding	12 × 39 turns, arc 28.2°, series, one bridge	12 × 39 turns, arc 29°, series, one bridge
L, R of the whole winding	2,974 µH, 6.15 Ω	2,973 µH, 6.15 Ω
AC: sources = load + copper + source R	2.334 = 0.297 + 1.156 + 0.881 W	1.462 = 0.570 + 0.562 + 0.331 W
AC efficiency	0.127	0.390

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	sine on A, cosine on B	drive phase evolved too
bank (kronspice, 4,700 µF, ~255 ms)	4.300 V, 43.45 mJ	6.097 V, 87.37 mJ
charging efficiency	0.0723	0.2262
ledger integration mismatch	142 µJ (2.2×10^{-4})	154 µJ (3.7×10^{-4})

The search fills every section with as many turns as fit and puts them in series into one bridge; separate or parallel bridges lose 4–50×, alternating sections cancel. Under sine/cosine the toroidal winding takes more power than Part 3's best outside coil (0.297 vs 0.252 W) at a third of its efficiency; left free to choose, the search drives the two windings in phase in the toroidal sense, where the same winding takes 0.570 W at 0.390 and banks 7× the coaxial coil's energy. A toroidal winding cannot use the rotation of the field vector that sine and cosine produce; it uses only the toroidal flux.

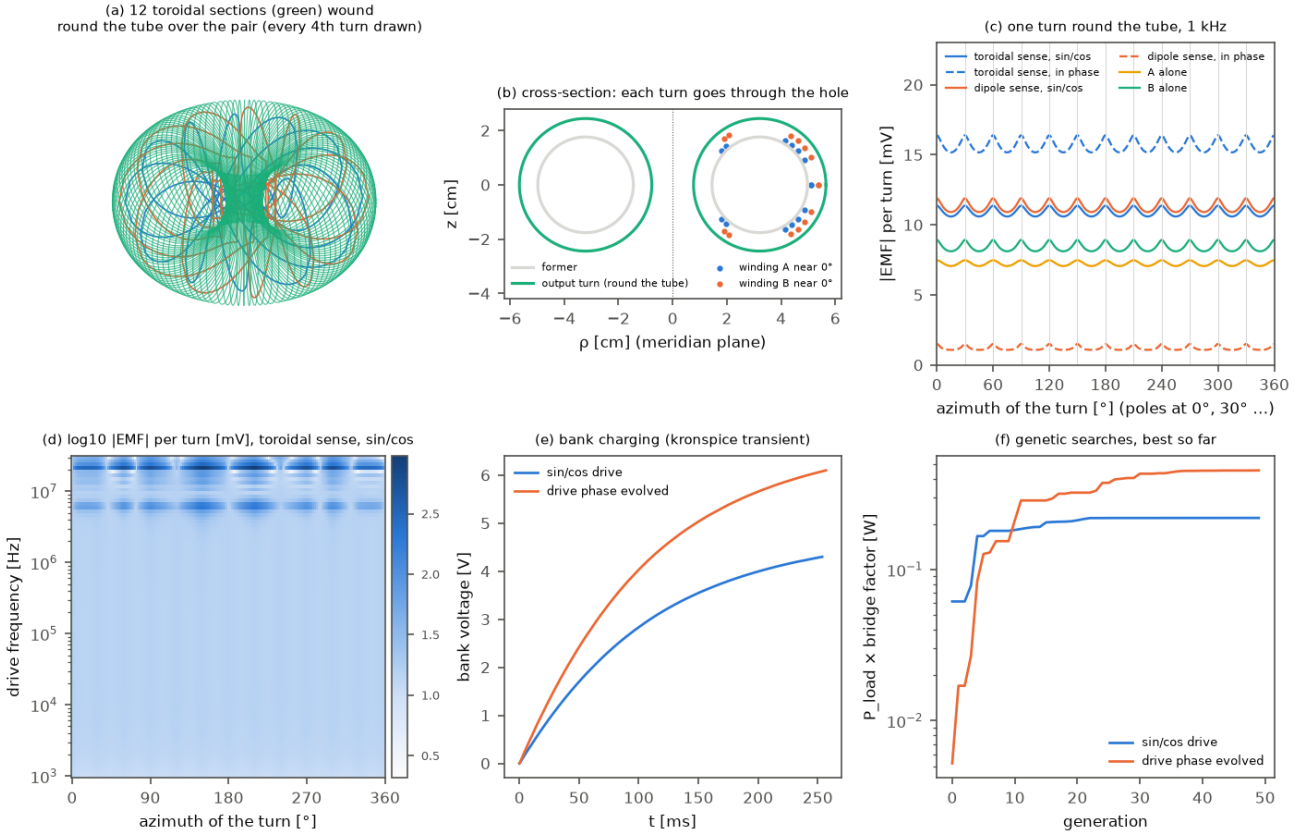


Figure 7. Part 3t: (a) the 12 sections on the pair, (b) a cross-section, (c) the EMF of one turn against angle at 1 kHz for every drive, (d) its (frequency, angle) map, (e) bank charging, (f) the searches.

Status of H43

claim	what the runs show	standard prediction
a “vortex” field	the vector turns at each point (ellipticity 0.09–0.11); the pattern stands still	two Biot–Savart fields a quarter cycle apart
the field accumulates	only by resonance, Q 76.4 vs 77.3, paid by the source; a tuned pair locks in phase	a series RLC tank
over-unity	every ledger closes; the bank holds 0.07–0.32 of the source energy (toroidal winding 0.07 under sine/cosine, 0.23 in phase)	transformer: output ≤ input
the (5, 12) path is special	it sets L, k and frequencies; 720° modes for (5, 12) and (2, 1) alike	a property of the connection
vortex-based mathematics	no equation given	—

Status: bounded. Computed under the standard model with closed books, the Rodin pair is a pair of coupled inductors (k = 0.39) and, near 2 MHz, a line laid on a (5, 12) knot. What would move the entry: a bench measurement in which the energy delivered to a load or bank exceeds the energy measured at the drive terminals, with every input counted. Bench test: the free-search winner (or the paper's coil with a coaxial secondary), a power

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analyser on both drive channels, and the bank's ½CV².

Approximations

- Turn count and gauge are not in the paper: 0.6 mm copper, 12 circuits per winding fitted to the measured centre field; each winding a round bundle on one path.
- Taut-wire path; crossings and curvature of a real winding idealized.
- Skin effect of an isolated wire, no proximity effect; air core.
- Sectional model with declared capacitances; results above the first self-resonance depend on them.
- Searches on 2 sections per circuit with a resistive load and a smooth bridge factor; the kronspice transient replaces them with real diodes.
- Radiation estimated (small loop), not modelled.
- Toroidal output winding: turns as circles at their azimuths, layers at their mean radius, leads twisted; no winding capacitance of the output winding.

Run it

```
go run ./examples/rodin_resonance
python3 examples/rodin_resonance/viz/plot_all.py
python3 examples/rodin_resonance/viz/plot_cycle.py
go test -short ./examples/rodin_resonance
```

Sources

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